# CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

**Sixth Semester of B. Tech. (CL) Examination**

**December 2013**

**CL 310 Irrigation & Hydraulic Structures**

**Date: 07.12.2013, Saturday** **Time: 01:30 p.m. To 04:30 p.m.** **Maximum Marks: 70**

***Instructions:***

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures with pencil only wherever required.
4. Use of scientific calculator is allowed.

## SECTION – I

**Q - 1 (a)** Define 'Irrigation' and discuss the factors which necessitate irrigation. **[03]**

**(b)** Enlist various methods of irrigation and discuss any one of them with their advantages and disadvantages. **[04]**

**Q - 2(a)** Define: Delta, Intensity of irrigation, Capacity factor, Field capacity. **[04]**

**(b)** The field capacity and permanent wilting point for the given soil are 35 and 15 per cent, respectively. Determine the storage capacity of the soil within root zone which may be taken as 80 cm. At a given time the soil moisture in the field is 20 per cent and farmer applies 25 cm of water. What part of this water would be wasted? Assume soil porosity as 40 per cent and relative density as 2.65 **[03]**

**(c)** Distinguish between 'Gross irrigation requirement' and 'Net irrigation requirement'. The consumptive use for a given crop is 90mm. Determine the field irrigation requirement if the effective rainfall and irrigation efficiency in the area are 15mm and 60% respectively. **[04]**

**(d)** Distinguish between 'Head regulator' and 'Cross regulator'. Discuss the functions served by the cross regulator. **[03]**

**OR**

**Q - 2(a)** What are the factors affecting duty? Suggest suitable methods of improving duty. **[03]**

**(b)** What is water application efficiency? The water application efficiency is very low in India. Suggest the ways and means to improve water application efficiency. **[03]**

**(c)** What are the different types of escapes? Explain the importance of surplus water escape in canal regulation. **[03]**

**(d)** For an irrigation project, compute the reservoir capacity with the following data: **[05]**

C. C. A = 40,000 ha. , Canal Losses = 13% ,

Reservoir Losses = 10%, Dead storage = 100 ha-m.

| Crop | Intensity of Irrigation (%) | Duty at canal head (ha/cumecs) | Base period (days) |
|---|---|---|---|
| Paddy | 35 | 850 | 120 |
| Sugarcan | 20 | 750 | 365 |
| Tobacco | 30 | 900 | 200 |
| Wheat | 40 | 1150 | 110 |

Q - 3(a) State different components of a diversion headwork and explain the functions of a divide wall [04]

(b) Explain the following terms: [04]

(i) Weighted creep length (ii) SSHGL (iii) Safe exit gradient (iv) Barrage

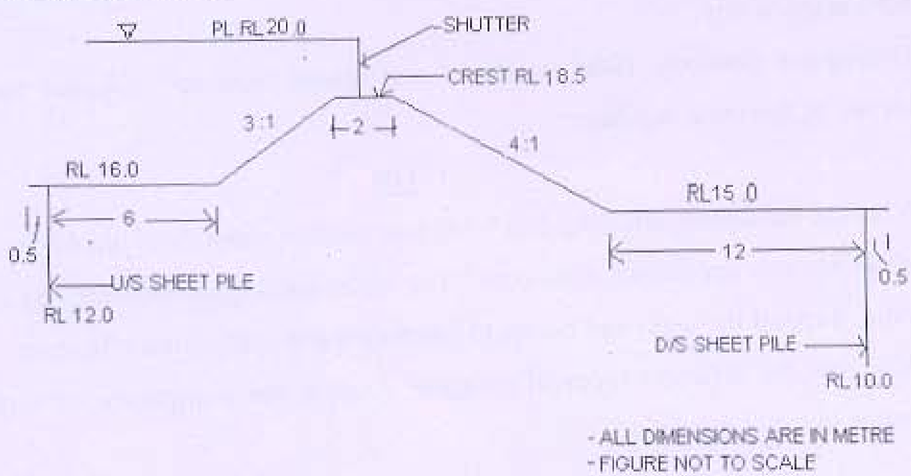
(c) What are the causes of failures of weir founded on permeable soils? Suggest the remedial measures. [04]

(d) Differentiate between 'super passage' and 'canal siphon', with neat sketches. [02]

**OR**

Q - 3 (a) Explain Bligh's creep theory for the design of weirs on permeable foundation. Also, discuss its limitations. [04]

(b) Obtain the residual uplift pressure heads at the key points for the weir shown in the Figure, using Khosla's theory. Also, check the safety of the weir against piping, if the safe exit gradient is 1/6. [08]




(c) Explain different types of 'Aqueduct' with neat sketches. [02]

## SECTION – II

Q – 4(a) Write function of the following galleries: [04]

Drainage gallery, Grouting gallery, Inspection gallery, Gate gallery

(b) Where the spillway is located? Enlist factors affecting capacity of a spillway. [03]

Q - 5 (a) Define: Canal capacity and Full supply level of canal [02]

(b) Design an unlined irrigation channel for non-alluvial soil using Chezy's formula for data given below: [05]

Q = 45 cumecs, Mean velocity = 0.9 m/sec, B = 5D, Side slope = 1:1, K = 1.2.

(c) Illustrate with neat sketches the following parts of an earthen dam and state their functions briefly: [07]

(i) Rock toe (ii) Horizontal drainage blanket (iii) Cut-off (iv) Rip-rap

**OR**

Q - 5 (a) Classify the canals by different ways. [02]

(b) Distinguish between high and low gravity dam with neat sketches. [03]

Find the limiting height of the low gravity dam with the following data: allowable compressive strength of concrete = 300 N/cm$^2$ and specific gravity of concrete is 2.41.

(c) Distinguish between elementary profile and practical profile with neat sketch of each. [04]

(d) Design a trapezoidal shaped concrete lined canal to carry a discharge of 100 cumecs at a slope of 25 cm/km. the side slope of channel is 1.5:1. The value of N is 0.016, limiting velocity = 1.5 m/sec. [05]

Q - 6 (a) Sketch an ogee profile and mark in it the different zones. [02]

(b) What is a 'chute spillway'? Where it is preferred to ogee and other types of spillways? [03]

(c) Distinguish between Super passage and siphon super passage. [04]

(d) Discuss the various causes of failure of embankment dam. [05]

**OR**

Q - 6 (a) Distinguish between Aqueduct and siphon aqueduct. [04]

(b) What is meant by an 'energy dissipator'? Discuss briefly the various types of energy dissipators that are used for energy dissipation below overflow spillways, under different relative positions of T.W.C and J.H.C. [05]

(c) Describe with neat sketches the various methods adopted for controlling seepage through the body of earthen dam. [05]

*******